



IILM UNIVERSITY

School of Computer Science and Engineering | Greater Noida, U.P.

INTERNSHIP PRESENTATION

A Web Development Internship at Thiranex: Building Accessible, Responsive & Full-Stack JavaScript Web Applications

Personal Portfolio • To-Do List • Weather Dashboard • ShopSphere Capstone

STUDENT NAME

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PROGRAM & SEMESTER

B.Tech CSE, 5th Semester

ORGANIZATION

Thiranex



Batch 2024–2028

What This Presentation Covers

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Organization Profile

Thiranex — role, mode, and internship duration

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Objectives & Scope

Skills targeted across the internship

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Technologies Used

The stack applied across all four projects

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The Four Projects

Portfolio, To-Do List, Weather Dashboard, ShopSphere

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Methodology & Outcomes

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Conclusion

Key takeaways from the internship

About Thiranex

Thiranex is a skill-development and future-tech organization offering remote, project-based internships in web development. Rather than theoretical modules, interns are given real, progressively harder build tasks under industry mentorship, with periodic performance review.

Role	Intern – Web Development
Mode	Remote / Project-Based
Duration	31 Jul 2026 – 30 Aug 2026
Intern ID	THX-JUL3126-333



Internship at a Glance

4

Web Applications Built

1

Month, Remote & Project-Based

10+

Core Web Technologies Applied

2

Live Public Deployments

The internship progressed from semantic, accessible front-end foundations through client-side application logic and API integration, culminating in a deployed full-stack capstone.

Objectives of the Internship

- 1 Build a semantic, accessible, WCAG-compliant multi-page portfolio
- 2 Apply advanced CSS3 (Grid, Flexbox, custom properties) for full responsiveness
- 3 Master JavaScript DOM logic, state management, and CRUD operations
- 4 Persist application state across sessions using browser storage
- 5 Consume live REST APIs using `fetch()` and `async/await`
- 6 Handle JSON data, loading states, and error conditions gracefully
- 7 Understand modular, component-based frontend architecture (React)
- 8 Deploy a production-ready application to a public hosting platform

Technologies & Tools Used

Markup & Styling

HTML5 (Semantic)

ARIA / WCAG

CSS3, Grid & Flexbox

Custom Properties / Theming

Application Logic

JavaScript ES6+

DOM API & Events

localStorage

React 18

Data & APIs

fetch() / async-await

REST APIs (OpenWeatherMap,
FakeStore)

JSON Processing

Error Handling

Delivery

Client-Side Routing

Git & Google Drive

Netlify / Lovable Hosting

Production Builds

PROJECT 1

Personal Portfolio Website

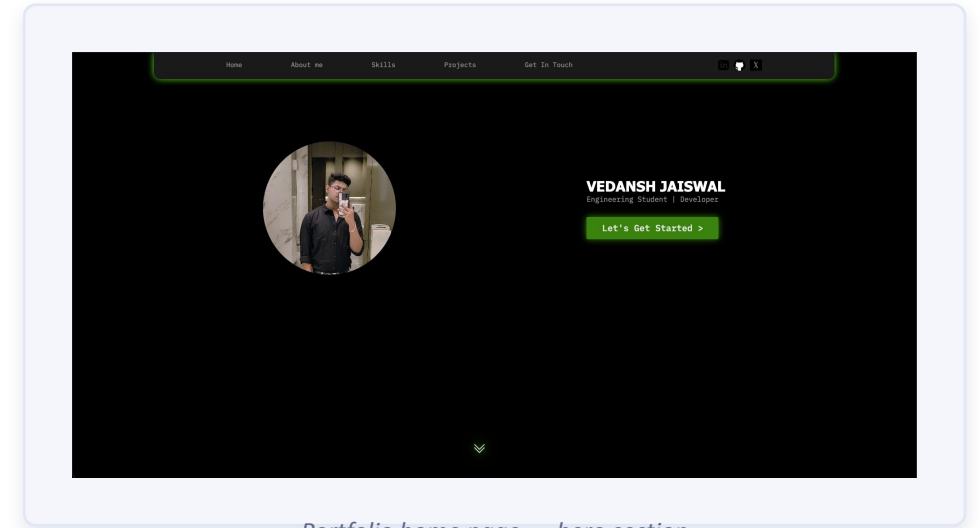
HTML5 Semantic Structure & Accessibility • Advanced CSS3 & Responsive Architecture

Built a multi-page personal portfolio adhering strictly to modern semantic HTML5 standards and WCAG accessibility guidelines, then transformed it into a fully responsive design using advanced CSS3 layout systems.

KEY FEATURES

- ◆ Semantic tags (header, nav, main, section, article, footer) throughout
- ◆ ARIA labels/roles and an accessible, tab-navigable contact form
- ◆ SEO-optimized meta tags
- ◆ CSS Grid + Flexbox layouts with mobile-first media queries
- ◆ Custom-property-driven light/dark theme

Live: vedanshportfolioo.netlify.app



Portfolio home page — hero section

To-Do List Application

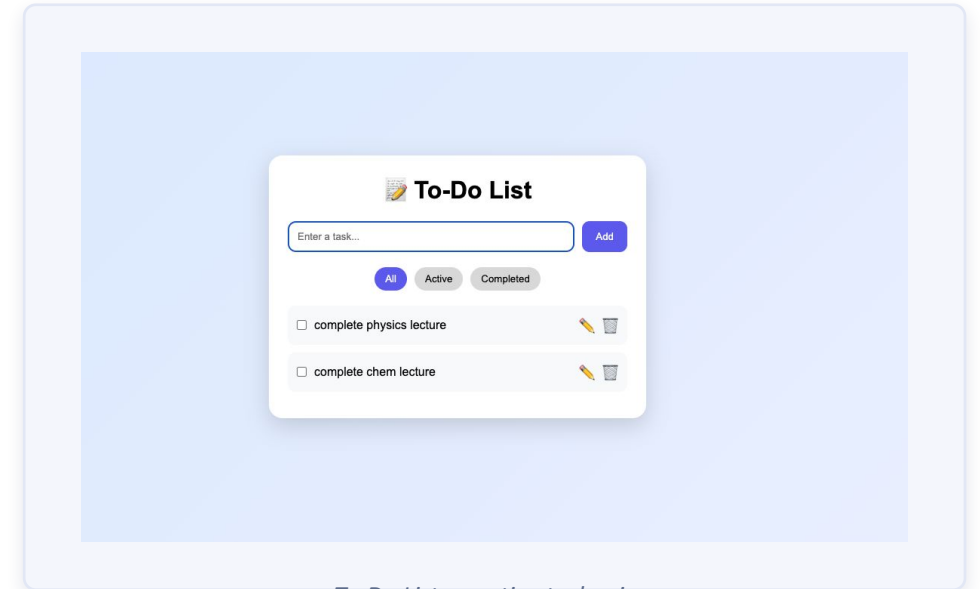
JavaScript Logic & State Management

Developed an interactive, client-side task manager to master DOM manipulation, event handling, and local data persistence — with full CRUD support and state that survives a page reload.

KEY FEATURES

- ◆ Full CRUD: create, read, update, delete tasks
- ◆ Filtering by All / Active / Completed
- ◆ Persistent storage via window.localStorage + JSON serialization
- ◆ Dynamically created DOM elements with delegated event listeners
- ◆ Live count of remaining tasks

Files: [Google Drive \(project folder\)](#)



To-Do List — active tasks view

Weather Dashboard

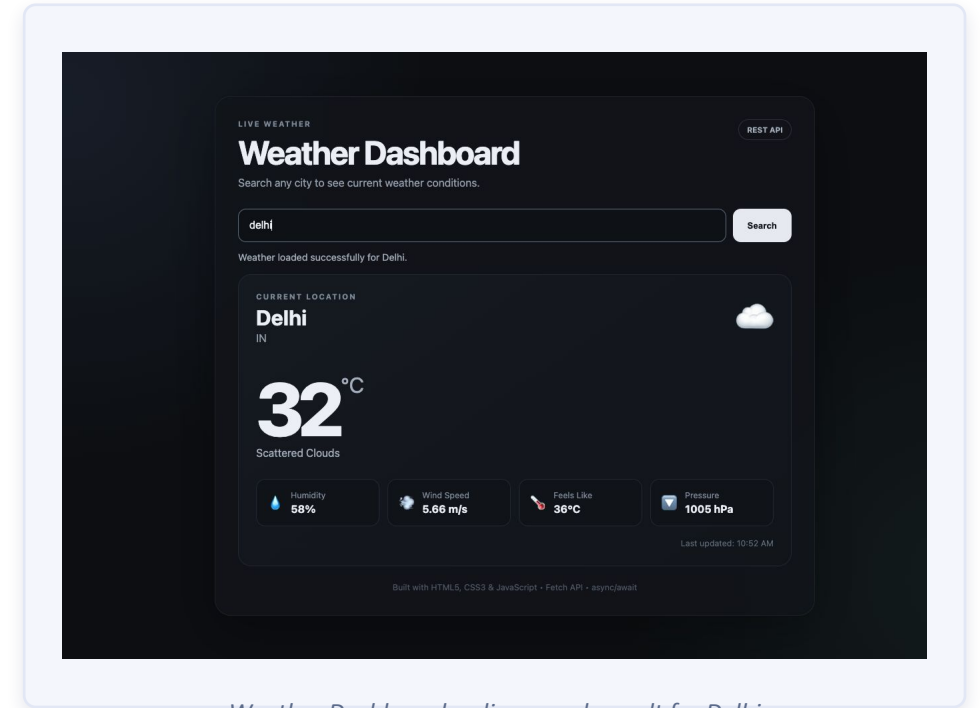
Asynchronous JavaScript & RESTful APIs

Built a real-time weather dashboard that fetches and renders live JSON data from a public REST API, using modern async/await syntax and robust error handling for a smooth user experience.

KEY FEATURES

- ◆ Live data via Fetch API + async/await
- ◆ City search with dynamic result rendering
- ◆ Temperature, humidity, wind speed, and pressure metrics
- ◆ Comprehensive error handling (invalid city, network failure)
- ◆ Responsive layout across devices

Files: [Google Drive \(project folder\)](#)



Weather Dashboard — live search result for Delhi

ShopSphere

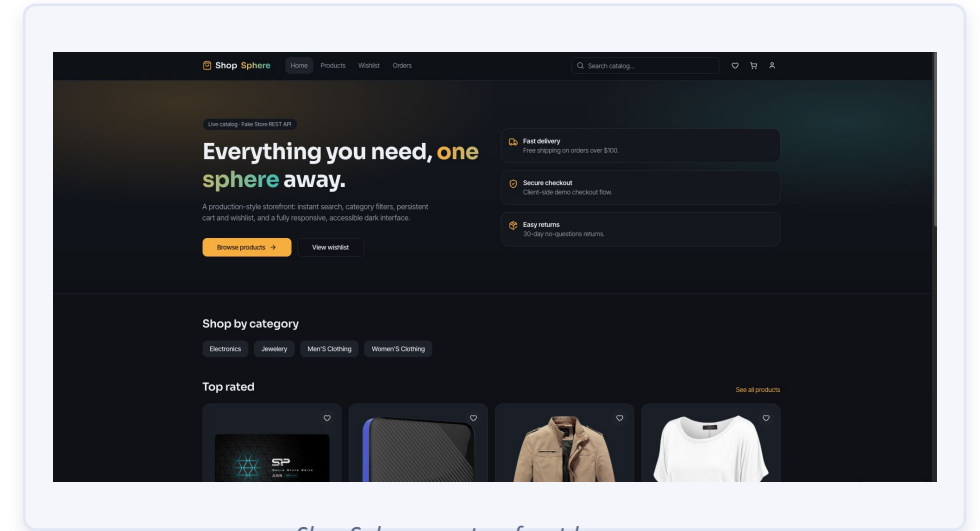
Full-Stack Deployment & Project Architecture

A production-style e-commerce product catalog that combines all prior learning: modular React architecture, client-side routing, state management, and a live public deployment.

KEY FEATURES

- ◆ Modular, component-based React architecture
- ◆ Client-side routing across catalog, product, cart, and account pages
- ◆ Product search, category filters, cart & wishlist (localStorage)
- ◆ REST API-driven product data with loading/error states
- ◆ Optimized production build, deployed live

Live: shopsphere-capstone.lovable.app



ShopSphere — storefront home page

Development Methodology



Learning Outcomes



HTML5 & Accessibility

Semantic markup, ARIA, SEO, accessible forms



CSS3 & Responsive Design

Grid, Flexbox, media queries, theming



JavaScript Fundamentals

Data structures, functions, event handling, DOM



State Management

App state, UI rendering, browser persistence



Web APIs

HTTP requests, REST APIs, JSON, async/await



Deployment

Production builds, asset optimization, live hosting

Challenges & Solutions

Accessibility compliance

→ Semantic HTML5 + ARIA labels; fully tab-navigable forms

Pixel-perfect responsiveness

→ CSS Grid for layout, Flexbox for alignment, mobile-first queries

Syncing UI with state

→ Central JS data structure; UI re-rendered on every state change

Data loss on refresh

→ localStorage with JSON serialization

Unreliable API requests

→ async/await + try/catch + loading & error states

Maintaining larger apps

→ Modular components, pages, and services (React)

Internship Completion Certificate



Conclusion

The Web Development internship at Thiranex provided practical, hands-on exposure to building interactive, accessible, and production-ready web applications — from a semantic, responsive portfolio through stateful JavaScript applications and API integration, culminating in a deployed full-stack capstone.

- ✦ **Bridged the gap between academic concepts and real application development**
- ✦ **Gained end-to-end experience: design → build → test → deploy**
- ✦ **Built and shipped four distinct, verifiable web applications**

Deliverables

- Personal Portfolio
- To-Do List
- Weather Dashboard
- ShopSphere Capstone

Thank You

Questions & Discussion